

Shubha Swarnim Singh

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Education

The College of Idaho, Computer Science and Business Administration

August 2022 – May 2026

- Gibson Honors Student, Dean's List, Honor's GPA: 3.77/4.0

Experience

Data Science and Analyst Intern, Albertsons Companies

June 2025 – August 2025

- Built supervised classification models in Python using Random Forest and XGBoost to predict store profitability across 2,200+ locations, integrating shrink, PWPS, and regional features.
- Developed ML-driven profitability pipeline analysis with Tableau and Power BI visualizations, supporting \$700M+ sales decisions and guiding CAPEX planning in the merchandising department.
- Automated store segmentation and breakeven analysis, reducing manual analysis time by 87.5%.

IT Intern, College of Idaho

Dec 2022 – May 2023

- Deployed Multi-Factor Authentication (MFA) across 1,000+ , achieving 90% adoption with secure token validation.
- Managed 1M+ records in SQL database; wrote SQL scripts ensuring data integrity and supporting faculty research.

Alumni & Events Technician, College of Idaho

Dec 2022 – Present

- Supported alumni fundraising events (\$1.1M and \$1.3M raised), managing outreach and database-driven targeting.
- Enhanced alumni database, improving engagement and retention by 20%.

Projects

Senate Voting System

GitHub

- Engineered a secure full-stack online voting system using Node.js, Express, SQLite, and WebSockets to digitize Senate voting for a 20-member student body, replacing legacy manual processes.
- Built real-time vote visualizations with <50ms latency using WebSockets and D3.js for instant, transparent updates.
- Added role-based access and authentication, enabling secret ballots and cutting groupthink by over 90%.

In-Memory Search Engine

GitHub

- Developed a high-performance Python search engine indexing 3,000+ .txt files (500MB) with <10ms latency.
- Implemented recursive directory crawling and created a word-based inverted index, reducing search time by over 90% compared to linear file scans.

Investment Calculator

GitHub

- Built Python tool leveraging Yahoo Finance API + sentiment analysis to analyze 2,500+ NASDAQ stocks.
- Designed an algorithm to support consistent, data-driven investment decisions.

Leadership

- ASCI Treasurer (2024/25) & Senator (2022-2024)**: Managed \$100K budget, ensuring effective fund distribution and financial transparency; previously served as a Senator.
- Consul, Sigma Chi (2025-Present)**: Led the fraternity, overseeing operations and brother development.
- Resident Assistant (2022-Present)**: Built inclusive community, resolved conflicts, and mentored peers.
- President, Boxing Club (2023-Present)**: Restarted and grew the club and built a supportive fitness community.

Technical Skills

Programming & Development: Python, JavaScript, SQL, Node.js, Express, Flask, HTML/CSS, Bootstrap, WebSocket

Data Science & Machine Learning: XGBoost, Random Forest, K-means, SHAP, Scikit-learn, Pandas, NumPy

Coursework: Data Structures, Algorithms, Machine Learning, Database Systems, Software Engineering, Programming Languages, Formal Automata